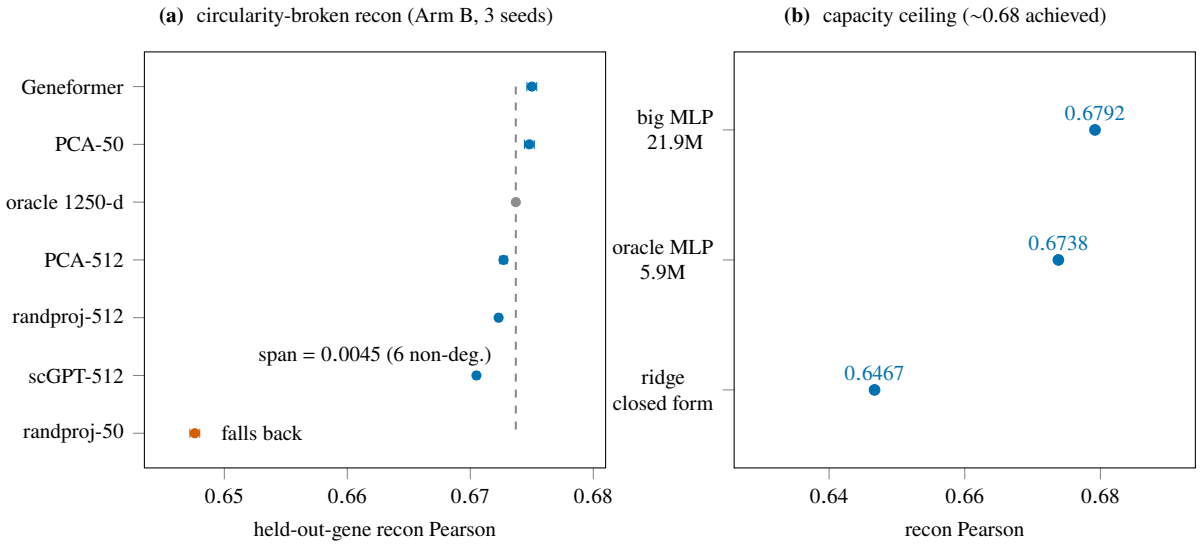

(a) circularity-broken recon (Arm B, 3 seeds)
Geneformer
PCA-50
oracle 1250-d
PCA-512
randproj-512
scGPT-512
randproj-50
span = 0.0045 (6 non-deg.)
falls back
0.65
0.66
0.67
0.68
held-out-gene recon Pearson
(b) capacity ceiling (~0.68 achieved)
big MLP
21.9M
oracle MLP
5.9M
ridge
closed form
0.6792
0.6738
0.6467
0.64
0.66
0.68
recon Pearson